

Course: Networked Control

Computer session 02: Decentralized and distributed control

Prof. Marcello Farina

Dipartimento di Elettronica, Informazione e Bioingegneria (DEIB)

Politecnico di Milano, Italy

`marcello.farina@polimi.it`

Outline

- Computation of the fixed modes.
- Computation of “structured” control gains for continuous and discrete-time systems
- Exercise: Coupled cart-stick balancers

Outline

- Computation of the fixed modes.
- Computation of “structured” control gains for continuous and discrete-time systems
- Exercise: Coupled cart-stick balancers

Computation of the fixed modes

Centralized fixed modes (CFM)

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ y(t) = Cx(t) \end{cases}$$

Definition: The CFMs are the elements of the set

$$\Lambda_{CFM} = \bigcap_{K_y \in \mathbb{R}^{m \times p}} \text{spec}(A + BK_y C)$$

Remark: They are the modes of the system which are

- uncontrollable, i.e., such that
$$\text{rank}([\lambda I - A \quad B]) < n, \text{ and/or}$$
- unobservable, i.e., such that
$$\text{rank} \left(\begin{bmatrix} \lambda I - A \\ C \end{bmatrix} \right) < n$$

Computation of the fixed modes

Centralized fixed modes (CFM)

- Modes can be placed in arbitrary locations using a dynamic output-feedback controller if and only if they are centralized «non-fixed».
- If a mode λ is a CFM, the discrete-time system (obtained by, e.g., sampling-and-hold discretization with sampling time h)

$$\begin{cases} x(k+1) = Fx(k) + Gu(k) \\ y(k) = Hx(k) \end{cases}$$

has a corresponding (discrete-time) centralized fixed mode in $e^{\lambda h}$.

Definition: The discrete-time CFMs are the elements of the set

$$\Lambda_{CFM}^{\text{discrete-time}} = \bigcap_{K_y \in \mathbb{R}^{m \times p}} \text{spec}(F + GK_yH)$$

Computation of the fixed modes

Centralized fixed modes (CFM) - computation

Continuous-time CFMs

1. Compute the spectrum of matrix A , i.e., $\text{spec}(A)$.
2. Define $\Lambda_{CFM} = \text{spec}(A)$.
3. Select the $K_y \in \mathbb{R}^{m \times p}$ randomly.
4. Compute $\text{spec}(A + BK_yC)$.
5. Set $\Lambda_{CFM} = \text{spec}(A + BK_yC) \cap \Lambda_{CFM}$.
6. Repeat procedure from step 3 “a number of times”.

Discrete-time CFMs

1. Compute the spectrum of matrix F , i.e., $\text{spec}(F)$.
2. Define $\Lambda_{CFM}^{\text{discrete-time}} = \text{spec}(F)$.
3. Select the matrix $K_y \in \mathbb{R}^{m \times p}$ randomly.
4. Compute $\text{spec}(F + GK_yH)$.
5. Set $\Lambda_{CFM}^{\text{discrete-time}} = \text{spec}(F + GK_yH) \cap \Lambda_{CFM}^{\text{discrete-time}}$.

Computation of the fixed modes

Decentralized fixed modes (DFM)

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ y(t) = Cx(t) \end{cases}$$

where $Bu(t) = B_1u_1(t) + \dots + B_Nu_N(t)$

$$y_i(t) = C_i x(t), i = 1, \dots, N$$

Definition: The DFM's are the elements of the set

$$\Lambda_{DFM} = \bigcap_{K_y^d \in \mathcal{K}_y^d} \text{spec}(A + BK_y^d C)$$

Set of the block diagonal gain matrices with the suitable size, i.e.,

$$K_y^d = \text{diag}(K_y^1, \dots, K_y^N)$$

where

$$K_y^i \in \mathbb{R}^{m_i \times p_i}$$

Computation of the fixed modes

Decentralized fixed modes (DFM)

- Modes can be placed in arbitrary locations using a DECENTRALIZED dynamic output-feedback LINEAR and TIME-INVARIANT controller if and only if they are decentralized «non-fixed».
- If a mode λ is a DFM, the discrete-time system (obtained by, e.g., sampling-and-hold discretization with sampling time h)

$$\begin{cases} x(k+1) = Fx(k) + Gu(k) \\ y(k) = Hx(k) \end{cases}$$

DOES NOT NECESSARILY have a corresponding (discrete-time) decentralized fixed mode in $e^{\lambda h}$.

Definition: The discrete-time DFMs are the elements of the set

$$\Lambda_{DFM}^{\text{discrete-time}} = \bigcap_{K_y^d \in \mathcal{K}_y^d} \text{spec}(F + GK_y^d H)$$

Computation of the fixed modes

Decentralized fixed modes (DFM) - computation

Continuous-time DFMs

1. Compute the spectrum of matrix A , i.e., $\text{spec}(A)$.
2. Define $\Lambda_{DFM} = \text{spec}(A)$.
3. Select the block-diagonal matrix $K_y^d \in \mathcal{K}_y^d$ randomly.
4. Compute $\text{spec}(A + BK_y^d C)$.
5. Set $\Lambda_{DFM} = \text{spec}(A + BK_y^d C) \cap \Lambda_{DFM}$.
6. Repeat procedure from step 3 “a number of times”.

Discrete-time DFMs

1. Compute the spectrum of matrix F , i.e., $\text{spec}(F)$.
2. Define $\Lambda_{DFM}^{\text{discrete-time}} = \text{spec}(F)$.
3. Select the block-diagonal matrix $K_y^d \in \mathcal{K}_y^d$ randomly.
4. Compute $\text{spec}(F + GK_y^d H)$.
5. Set $\Lambda_{DFM}^{\text{discrete-time}} = \text{spec}(F + GK_y^d H) \cap \Lambda_{DFM}^{\text{discrete-time}}$.

Computation of the fixed modes

Distributed fixed modes (DiFM)

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) & \text{where } Bu(t) = B_1u_1(t) + \dots + B_Nu_N(t) \\ y(t) = Cx(t) & y_i(t) = C_ix(t), i = 1, \dots, N \end{cases}$$

Information structure constraint

- $\mathcal{N}_i := \{j \in \mathcal{I} : y_j \text{ is available to controller } \mathcal{C}_i\}$
- $\mathcal{N} := \{\mathcal{N}_1, \dots, \mathcal{N}_N\}$
- $\mathcal{K}_y^{\mathcal{N}}$: set of the gain matrices with the suitable structure, i.e.,

$$K_y = \begin{bmatrix} K_y^{11} & \dots & K_y^{1N} \\ \vdots & \ddots & \vdots \\ K_y^{N1} & \dots & K_y^{NN} \end{bmatrix}$$

such that, for all $i = 1, \dots, N$, $K_y^{ij} = 0$ if $j \notin \mathcal{N}_i$.

Definition: The DiFMs are the elements of the set

$$\Lambda_{DiFM} = \bigcap_{K_y \in \mathcal{K}_y^{\mathcal{N}}} \text{spec}(A + BK_yC)$$

Computation of the fixed modes

Distributed fixed modes (DiFM)

- Modes can be placed in arbitrary locations using a DISTRIBUTED dynamic output-feedback LINEAR and TIME-INVARIANT controller if and only if they are distributed «non-fixed».
- If a mode λ is a DiFM, the discrete-time system (obtained by, e.g., sampling-and-hold discretization with sampling time h)

$$\begin{cases} x(k+1) = Fx(k) + Gu(k) \\ y(k) = Hx(k) \end{cases}$$

DOES NOT NECESSARILY have a corresponding (discrete-time) distributed fixed mode in $e^{\lambda h}$.

Definition: The discrete-time DiFMs are the elements of the set

$$\Lambda_{DiFM}^{\text{discrete-time}} = \bigcap_{K_y \in \mathcal{K}_y^{\mathcal{N}}} \text{spec}(F + GK_yH)$$

Computation of the fixed modes

Distributed fixed modes (DiFM) - computation

Continuous-time DiFMs

1. Compute the spectrum of matrix A , i.e., $\text{spec}(A)$.
2. Define $\Lambda_{DiFM} = \text{spec}(A)$.
3. Select the structured matrix $K_y \in \mathcal{K}_y^{\mathcal{N}}$ randomly.
4. Compute $\text{spec}(A + BK_yC)$.
5. Set $\Lambda_{DiFM} = \text{spec}(A +$

Discrete-time DiFMs

1. Compute the spectrum of matrix F , i.e., $\text{spec}(F)$.
2. Define $\Lambda_{DiFM}^{\text{discrete-time}} = \text{spec}(F)$.
3. Select the structured matrix $K_y \in \mathcal{K}_y^{\mathcal{N}}$ randomly.
4. Compute $\text{spec}(F + GK_yH)$.
5. Set $\Lambda_{DiFM}^{\text{discrete-time}} = \text{spec}(F +$

Computation of the fixed modes (code)

1. Define the “structure” of the controller using the matrix $\text{ContStruc} \in \mathbb{R}^{N \times N}$: its (i, j) -th element is

- $= 0$ if $K_y^{ij} = 0$;
- $= 1$ if $K_y^{ij} \neq 0$.

For example:

- Centralized control (or distributed control with all-to-all communication):

```
>> ContStruc=ones(N,N)
```

- Decentralized control:

```
>> ContStruc=diag(ones(N,1));
```

- Distributed control (Bidirectional star topology), e.g., with $N=4$:

```
>> ContStruc=[1 1 1 1
               1 1 0 0
               1 0 1 0
               1 0 0 1];
```

Computation of the fixed modes (code)

2. Initialize the set of fixed modes (rounded for avoiding the miss detection of fixed modes for numerical reasons):

```
>> FM=round(eig(A),rounding decimal);
```

3. Generation of a random matrix K compatible with the information exchange structure specified by the matrix ContStruc

```
>> K=zeros(m_tot,p_tot);  
    m_inc=0;  
    for i=1:N  
        p_inc=0;  
        for j=1:N  
            if ContStruc(i,j)~0  
  
K(m_inc+1:m_inc+m(i),p_inc+1:p_inc+p(j))=randn(m(i),p(  
j));  
  
                end  
                p_inc=p_inc+p(j);  
            end  
            m_inc=m_inc+m(i);  
        end
```

Computation of the fixed modes (code)

4. Compute the eigenvalues of $A+BKC$ (rounded for avoiding the miss detection of fixed modes for numerical reasons):

```
>> eigABKC=round(eig(A+B*K*C),rounding decimal);
```

5. Intersect FM and eigABKC:

```
>> FM=intersect(FM,eigABKC);
```

6. Repeat from step 3.

Outline

- Computation of the fixed modes.
- Computation of “structured” control gains for continuous and discrete-time systems
- Exercise: Coupled cart-stick balancers

Computation of “structured” control gains with LMIs (state-feedback) for continuous-time control

$$\dot{x}(t) = Ax(t) + B_1u_1(t) + \cdots + B_Nu_N(t) \quad x = \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix}, x_i \in \mathbb{R}^{n_i}$$

Information structure constraints

- $\mathcal{N}_i := \{j \in \mathcal{I} : x_j \text{ is available to controller } \mathcal{C}_i\}$
- $\mathcal{N} := \{\mathcal{N}_1, \dots, \mathcal{N}_N\}$
- K : gain matrix with the suitable structure, i.e.,
$$K = \begin{bmatrix} K^{11} & \dots & K^{1N} \\ \vdots & \ddots & \vdots \\ K^{N1} & \dots & K^{NN} \end{bmatrix}$$
such that, for all $i = 1, \dots, N$, $K^{ij} = 0$ if $j \notin \mathcal{N}_i$.

Computation of “structured” control gains with LMIs (state-feedback) for continuous-time control - code

1. Define the “structure” of the controller using the matrix $\text{ContStruc} \in \mathbb{R}^{N \times N}$: its (i, j) -th element is

- $= 0$ if $K_y^{ij} = 0$;
- $= 1$ if $K_y^{ij} \neq 0$.

For example:

- Centralized control (or distributed control with all-to-all communication):

```
>> ContStruc=ones(N,N)
```

- Decentralized control:

```
>> ContStruc=diag(ones(N,1));
```

- Distributed control (Bidirectional star topology), e.g., with $N=4$:

```
>> ContStruc=[1 1 1 1
              1 1 0 0
              1 0 1 0
              1 0 0 1];
```

Computation of “structured” control gains with LMIs (state-feedback) for continuous-time control - code

2. Generation of matrices L and Y compatible with the information exchange structure specified by the matrix $ContStruc$ (see later).

3. Definition of the LMI constraint (e.g., for stability)

```
>> LMIconstr=[Y*A'+A*Y+B*L+L'*B'<-1e-  
2*eye(n)]+[Y>1e-2*eye(n)];
```

4. Solve the LMI feasibility problem (a cost can be added)

```
>> J=optimize(LMIconstr);
```

5. Compute the control gain

```
>> L=double(L);  
>> Y=double(Y);  
>> K=L/Y;
```

6. The control action is

$$u(t) = Kx(t)$$

Computation of “structured” control gains with LMIs (state-feedback) for discrete-time control

$$x(k+1) = Fx(k) + G_1u_1(k) + \cdots + G_Nu_N(k) \quad x = \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix}, x_i \in \mathbb{R}^{n_i}$$

Information structure constraints

- $\mathcal{N}_i := \{j \in \mathcal{I} : x_j \text{ is available to controller } \mathcal{C}_i\}$
- $\mathcal{N} := \{\mathcal{N}_1, \dots, \mathcal{N}_N\}$
- K : gain matrix with the suitable structure, i.e.,
$$K = \begin{bmatrix} K^{11} & \cdots & K^{1N} \\ \vdots & \ddots & \vdots \\ K^{N1} & \cdots & K^{NN} \end{bmatrix}$$
such that, for all $i = 1, \dots, N$, $K^{ij} = 0$ if $j \notin \mathcal{N}_i$.

Computation of “structured” control gains with LMIs (state-feedback) for discrete-time control - code

1. Define the “structure” of the controller using the matrix $\text{ContStruc} \in \mathbb{R}^{N \times N}$: its (i, j) -th element is

- $= 0$ if $K_y^{ij} = 0$;
- $= 1$ if $K_y^{ij} \neq 0$.

For example:

- Centralized control:

```
>> ContStruc=ones(N,N)
```

- Decentralized control:

```
>> ContStruc=diag(ones(N,1));
```

- Distributed control (Bidirectional star topology), e.g., with $N=4$:

```
>> ContStruc=[1 1 1 1
               1 1 0 0
               1 0 1 0
               1 0 0 1];
```

Computation of “structured” control gains with LMIs (state-feedback) for discrete-time control - code

2. Generation of matrices L and P compatible with the information exchange structure specified by the matrix $ContStruc$ (see later).

3. Definition of the LMI constraint (e.g., for stability)

```
>> LMIconstr=[ [P-F*P*F'-F*L'*G'-G*L*F', G*L;  
               L'*G'                        ,      P]>1e-2*eye(n*2)];
```

4. Solve the LMI feasibility problem (a cost can be added)

```
>> J=optimize(LMIconstr);
```

5. Compute the control gain

```
>> L=double(L);
```

```
>> P=double(P);
```

```
>> K=L/P;
```

6. The control action is

$$u(k) = Kx(k)$$

Computation of “structured” control gains with LMIs (state-feedback) for discrete-time control - code

Generation of matrices L and P (or, similarly, of Y) compatible with the information exchange structure specified by the matrix *ContStruc* (see later).

- If *ContStruc*=ones (N, N) [centralized or distributed all-to-all structure]:
>> $P = \text{sdpvar}(n);$
 $L = \text{sdpvar}(m, n);$
- If *ContStruc*≠ones (N, N) [decentralized/distributed structure]:
>> $P = [];$
 $L = \text{sdpvar}(m, n);$
 $\text{minc} = 0;$
 for $i = 1:N$
 $P = \text{blkdiag}(P, \text{sdpvar}(n_i(i)));$
 $n_{\text{inc}} = 0;$
 for $j = 1:N$
 if $\text{ContStruc}(i, j) == 0$
 $L(\text{minc}+1:\text{minc}+m(i), n_{\text{inc}}+1:n_{\text{inc}}+n_i(j))$
 $= \text{zeros}(m(i), n_i(j));$
 end
 $n_{\text{inc}} = n_{\text{inc}} + n_i(j);$
 end
 $\text{minc} = \text{minc} + m(i);$
 end
end

Outline

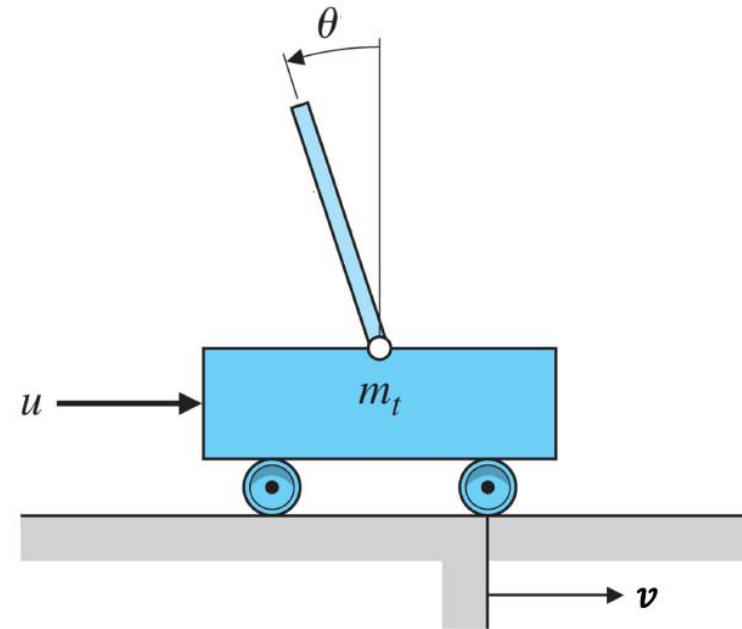
- Computation of the fixed modes.
- Computation of “structured” control gains for continuous and discrete-time systems
- Exercise: Coupled cart-stick balancers

Exercise: Coupled (through dumpers) cart-stick balancers

Isolated Cart-Stick Balancer (inverted pendulum)

Consider the cart-stick balancer system in the figure. Its model has three states, one input, and one output:

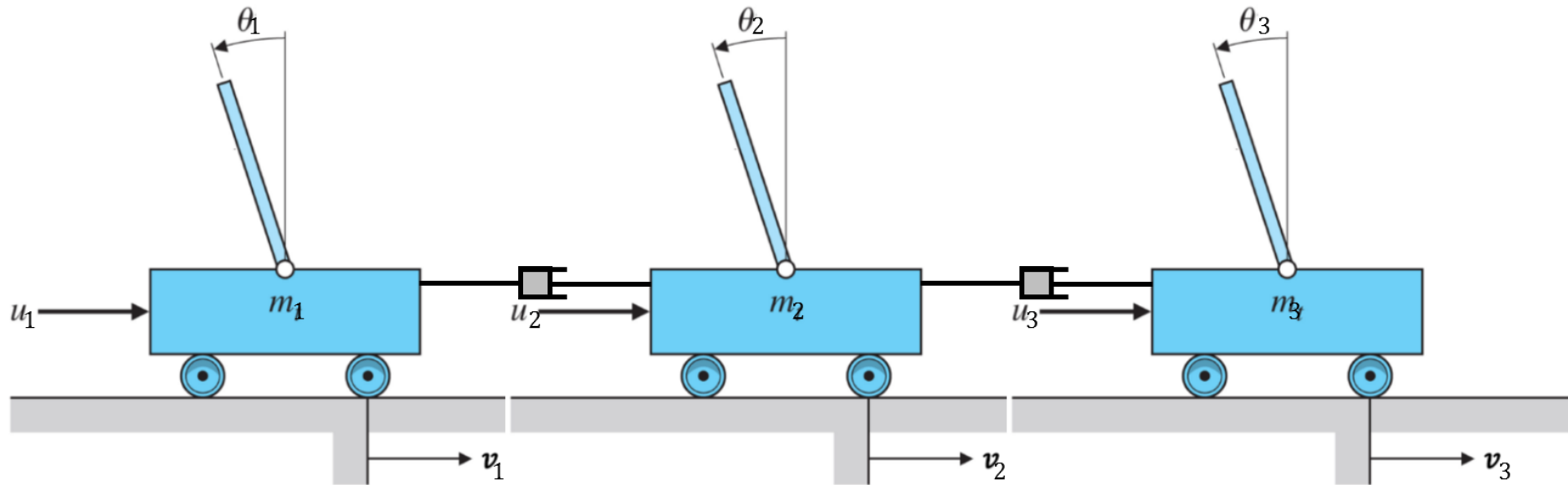
- $x_1: \theta/10$ – stick angle/10
- $x_2: \dot{\theta}/10$ – stick angular velocity/10
- $x_3 = v$ cart velocity
- u : voltage of the motor driving the wheels



The linearized model is

$$\dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 31.33 & 0 & 0.016 \\ -31.33 & 0 & -0.216 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ -0.649 \\ 8.649 \end{bmatrix} u$$

Exercise: Coupled (through dampers) cart-stick balancers



Each cart i has three states $(\theta_i/10, \dot{\theta}_i/10, v_i)$ and one input (u_i) . The output, for each cart, is

$$y_i = x_i = \begin{bmatrix} \theta_i/10 \\ \dot{\theta}_i/10 \\ v_i \end{bmatrix}$$

Control goal: maintain the sticks vertical, i.e., at $\theta_i = 0$ for all $i = 1, 2, 3$.

Exercise: Coupled (through dumpers) cart-stick balancers

For different values of 'N' (number of coupled carts), 'coupling' (number proportional to the dumping factor of the dumpers), 'h', perform the following actions.

1. Generate the system matrices (both continuous-time and discrete-time) using the function 'coupled_CSB.m'

```
>> help coupled_CSB
```

Defines the model of N coupled (through dumpers) Cart Stick Balancers

Inputs:

- N (number of subsystems)
- coupling (coupling value, e.g., 1,2,...)
- h (sampling time)

Outputs:

- A: CT system matrix
- B: CT input matrices (i.e., B{1},..., B{N} are the input matrices of the decomposed system, one for each channel)
- C: CT output matrices (i.e., C{1},..., C{N} are the output matrices of the decomposed system, one for each channel)
- F: DT system matrix
- G: DT input matrices (i.e., G{1},..., G{N} are the input matrices of the decomposed system, one for each channel)
- H: DT output matrices (i.e., H{1},..., H{N} are the output matrices of the decomposed system, one for each channel)

Exercise: Coupled (through dumpers) cart-stick balancers

2. For different “control structures” perform the following actions
 - a) Compute the continuous-time fixed modes
 - b) Compute the discrete-time fixed modes
 - c) Compute, if possible, the corresponding stabilizing CONTINUOUS-TIME control gain using the LMIs
 - d) Compute, if possible, the corresponding stabilizing DISCRETE-TIME control gain using the LMIs
 - e) Compute the closed-loop system trajectories (generated both in continuous-time and in discrete-time) starting from a common random initial condition.

SUGGESTION: the control structures can be defined by the matrix ‘ContStruc’. For example (when $N=3$):

- Centralized control: `>> ContStruc=ones(3,3)`
- Decentralized control: `>> ContStruc=diag(ones(3,1));`
- Distributed control (Bidirectional star topology):
`>> ContStruc=[1 1 1`
 `1 1 0`
 `1 0 1];`
- Distributed control (Bidirectional string topology):
`>> ContStruc=[1 1 0`
 `1 1 1`
 `0 1 1];`

Exercise: Coupled (through dumpers) cart-stick balancers

SUGGESTION: the fixed modes can be computed using the function

`'di_fixed_modes.m'`, e.g.,

```
>> help di_fixed_modes
```

```
Computes the fixed modes of the system, with reference to the  
control information structure specified by 'ContStruc'.
```

Inputs:

- A: system matrix - either continuous-time or discrete-time.
- B: input matrices - either continuous-time or discrete-time -
(i.e., B{1},..., B{N} are the input matrices of the
decomposed system, one for each channel).
- C: output matrices - either continuous-time or discrete-time -
(i.e., C{1},..., C{N} are the output matrices of the
decomposed system, one for each channel).
- N: number of subsystems.
- ContStruc: NxN matrix that specifies the information structure
constraints (ContStruc(i,j)=1 if communication is allowed
between channel j to channel i, ContStruc(i,j)=0 otherwise)
- rounding_n: the eigenvalues are rounded at the nth decimal place.

Output:

- Difm: vector containing the fixed modes (if empty, there are no
FMs)

Exercise: Coupled (through dumpers) cart-stick balancers

SUGGESTION: the corresponding stabilizing CONTINUOUS-TIME control gain using the LMIs can be computed using the function `'LMI_CT_DeDiCont.m'`, e.g.,

```
>> help LMI_CT_DeDiCont
```

Computes, using LMIs, the distributed "state feedback" control law for the continuous-time system, with reference to the control information structure specified by 'ContStruc'.

Inputs:

- A: system matrix.
- B: input matrices (i.e., $B\{1\}, \dots, B\{N\}$ are the input matrices of the decomposed system, one for each channel).
- C: output matrices (i.e., $C\{1\}, \dots, C\{N\}$ are the output matrices of the decomposed system, one for each channel, where $[C\{1\}', \dots, C\{N\}'] = I$).
- N: number of subsystems.
- ContStruc: $N \times N$ matrix that specifies the information structure constraints ($\text{ContStruc}(i,j)=1$ if communication is allowed between channel j to channel i , $\text{ContStruc}(i,j)=0$ otherwise).

Output:

- K: structured control gain
- rho: spectral abscissa of matrix $(A+B*K)$ - note that $[C\{1\}', \dots, C\{N\}'] = I$
- feas: feasibility of the LMI problem (=0 if yes)

Exercise: Coupled (through dumpers) cart-stick balancers

SUGGESTION: the corresponding stabilizing DISCRETE-TIME control gain using the LMIs can be computed using the function `'LMI_DT_DeDiCont.m'`, e.g.,

```
>> help LMI_DT_DeDiCont
```

Computes, using LMIs, the distributed "state feedback" control law for the discrete-time system, with reference to the control information structure specified by 'ContStruc'.

Inputs:

- F: system matrix.
- G: input matrices (i.e., $G\{1\}, \dots, G\{N\}$ are the input matrices of the decomposed system, one for each channel).
- H: output matrices (i.e., $H\{1\}, \dots, H\{N\}$ are the output matrices of the decomposed system, one for each channel, where $[H\{1\}', \dots, H\{N\}'] = I$).
- N: number of subsystems.
- ContStruc: $N \times N$ matrix that specifies the information structure constraints (ContStruc(i,j)=1 if communication is allowed between channel j to channel i, ContStruc(i,j)=0 otherwise).

Output:

- K: structured control gain
- rho: spectral radius of matrix $(F+G*K)$ - note that $[H\{1\}', \dots, H\{N\}'] = I$
- feas: feasibility of the LMI problem (=0 if yes)